# MZUZU UNIVERSITY

## FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

## DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

## COMPUTER AND COMMUNICATION TECHNOLOGY

## (SUPPLEMENTARY/DIFFERED)

## (ICT 1103)

**DATE: 13TH JUNE, 2017** **TIME ALLOWED: 2:30 HOURS**

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**INSTRUCTIONS**

a. There are Five (**5**) questions. **PLEASE CHECK.**
b. There are **two** pages. **PLEASE CHECK.**
c. Answer **ALL** questions.

## Question One

1. List **four** distinct parts of Information Technology systems in an organization. **(4 Marks)**
2. Mention and explain **four** impacts of Information Technology in an organization. **(8 Marks)**
3. What is the difference between Information Systems and Information Technology? **(4 Marks)**
4. Mention **four** primary issues in Computer Ethics. **(4 Marks)**
5. Mention and discuss **four** threats of computer and communication systems. **(8 Marks)**

## Question Two

1. Explain the following Data Communication Terminologies:
   i. Channel **(2 Marks)**
   ii. Bandwidth **(2 Marks)**
2. Mention **two** principle benefits of computer networks. **(2 Marks)**
3. Mention and discuss **three** classifications of computer networks. **(9 Marks)**
4. Define the term Network Topology. **(2 Marks)**
5. What is the difference between Logical and Physical topologies? **(2 Marks)**
6. Explain any **three** types of topologies in computer networks. **(9 Marks)**

## Question Three

Most organisations utilise a variety of information and usually have more than one information system and below is the list of those information systems.

1. Give **three** examples of each of the following:
   a. Executive Support Systems/Executive Information Systems **(3 Marks)**
   b. Management Information Systems **(3 Marks)**
   c. Knowledge Management Systems **(3 Marks)**
   d. Transaction Processing Systems **(3 Marks)**
   e. Office Automation Systems **(3 Marks)**
2. State **four** purposes of doing analysis and requirements capture in Systems Development Life Cycle. **(4 Marks)**

## Question Four

1. Mention **four** database functions. **(4 Marks)**
2. List down **four** features of the Database approach. **(4 Marks)**
3. Mention **three**properties of a File Processing Systems (file based system). **(3Marks)**

## Question Five

1. What are Numbering systems? **(2 Marks)**
2. What is the difference between Encoding and Decoding? **(2 Marks)**
3. Convert the following:
   i. $1010001011011110_2$ to Hexadecimal number systems **(2 Marks)**
   ii. $FF_{16}$ to binary number. **(2 Marks)**
   iii. $E5_{16}$ to octal number system. **(2 Marks)**
   iv. $231_{10}$ to octal number system. **(2 Marks)**
   v. $121336_8$ to hexadecimal number systems. **(2 Marks)**

# END OF QUESTION PAPER